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Short Communication

Secondary thermophilic microaerobic treatment in the anaerobic digestion of corn straw

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HIGHLIGHTS

- Secondary thermophilic microaerobic treatment (STMT) was carried out.
- STMT obviously improved the efficiency of the late fermentation stage.
- Corn straw with STMT obtained 28.45% higher methane yield.
- The VS removal efficiency after STMT improved for 29.43%.

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ABSTRACT

Thermophilic microaerobic pretreatment (TMP) has been proved to be an alternative pretreatment method during anaerobic digestion (AD) of corn straw. In this study, in order to improve the fermentation efficiency during late AD stage, improve the methane yield and volatile solid (VS) removal efficiency, a secondary thermophilic microaerobic treatment (STMT) was applied in the late AD stage of corn straw. Results showed STMT obviously improved the fermentation efficiency, methane yield and VS removal efficiency. The maximum methane yield and maximum VS removal efficiency were simultaneously obtained when the oxygen loads during STMT was 10 ml/g VS (VS of residual substrate). The maximum methane yield was 380.6 ml/g $VS_{substrate}$, which was 28.45% and 10.61% higher than those of untreated and once thermophilic microaerobic pretreated samples, respectively. The maximum VS removal efficiency was 81.85%, which was 29.43% and 17.23% higher than those of untreated and once thermophilic microaerobic pretreated samples, respectively.

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1. Introduction

Due to the increasing attentions to renewable energy, climate change and waste management, anaerobic digestion (AD) as an ideal way for the comprehensive utilization of organic waste has been widely employed (Sanscartier et al., 2012). Corn straw, which is rich in cellulose, hemicelluloses and lignin, is one of the largest residues from agriculture. It has a great potential for biogas production because of its abundance and high carbohydrate content (You et al., 2014). However, the strong interlinkage of cellulose, hemicelluloses and lignin with noncovalent forces and covalent cross-linkage gives a stable shape and structure to the plant (NIZAMI et al., 2009), which are the barriers of the enzymolysis during hydrolysis process. Therefore, the AD of corn straw is inefficient. Pretreatment is essential to decrease the structural and compositional barriers by breaking down lignin interference, reducing cellulose crystallinity and increasing surface area (Yu et al., 2014).

Recently, microaerobic pretreatment has been proved to be an efficient pretreatment method for the anaerobic digestion in several studies (Botheju et al., 2010; Charles et al., 2009; Fu et al., 2015; Mshandete et al., 2005). The term microaeration refers to the controlled introduction of small amounts of oxygen into an anaerobic biochemical process to enable both anaerobic and aerobic biological activities to occur within a single bioreactor (Lim et al., 2014). Compared to other methods, microaerobic pretreatment is more economy and environmental friendly, which just

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needs limited amount of oxygen (or air) supplied directly into the anaerobic digester or during a pretreatment step (Ramos and Fdz-Polanco, 2013).

In our earlier paper study on the effect of thermophilic microaerobic pretreatment on the anaerobic digestion of corn straw, the thermophilic microaerobic pretreatment before anaerobic digestion of corn straw at the oxygen loads of 5 ml/g $VS_{substrate}$ demonstrated a 16.24% higher methane yield (Fu et al., 2015). However, the VS removal efficiency at this condition was only 68.06%, which means there are still a relevant portion of corn straw that were not been digested. During anaerobic digestion of cellulosic substrate like corn straw, easily degradable compounds are firstly utilized at the early stage of AD, the residual recalcitrant compositions are mainly utilized at the late stage of fermentation, and thus the fermentation at the late stage is inefficient. In order to improve the fermentation efficiency during the late stage, a secondary thermophilic microaerobic treatment (STMT) was proposed in this study. The secondary thermophilic microaerobic treatment at the later period of anaerobic digestion when obvious decreases in daily methane yield were appeared could be a solution to improve the low fermentation efficiency during the late stage.

The objective of this study was to investigate the effect of the secondary thermophilic microaerobic treatment on the anaerobic digestion of corn straw. In addition, different oxygen loads during secondary thermophilic microaerobic treatment were also applied to investigate the effect of oxygen load on the performance of secondary thermophilic microaerobic treatment.

2. Methods

2.1. Substrate and inoculum

The substrate used in this study was corn straw, which was collected from corn field in Pingdu (Shandong Province, China). The corn straw was chopped and sieved to a size of less than 0.5 cm by a 0.5 cm sieve. The total solid (TS) and volatile solid (VS) of corn straw were 92.44% and 93.44% (based on TS), respectively. The C/N and C/H ratio are 28.97 and 7.73, respectively (The Element contents of corn straw were quantified with an elemental analyzer, Vario EL Cube). The proportions of cellulose, hemicelluloses and lignin are 45.43%, 22.73% and 10.79% (% of TS), respectively.

Inoculum used in this study was biogas slurry, which was collected from a 500 m^3 size of biogas plant (Qingdao, Shandong Province, China). The TS and VS contents of the biogas slurry are 6.10% and 73.05% (% of TS), respectively.

2.2. Thermophilic microaerobic pretreatment (TMP) of corn straw

The TMP stage total contained 10 bottles, the bottles were divided into 5 groups (marked as R0, R, RS1, RS2 and RS3). R0 was used for testing the VS removal efficiency before STMT, R was used for testing the biogas production from once thermophilic microaerobic pretreated corn straw, RS1, RS2 and RS3 were used for testing the effect of STMT on the anaerobic digestion of corn straw. During TMP stage, 5.77 g corn straw and 22 ml inocula were mixed in bottles, and then deionized water was added to reach a total volume of 0.1 L. Each bottle was flushed with N_2 for 5 min to replace the air, and then the bottles were closed with rubber stoppers. Then 25 ml oxygen at atmospheric pressure was injected to the bottles with a syringe to reach the oxygen loads of 5 ml/g VS substrate, which was regarded as the optimized oxygen loads in previous study (Fu et al., 2015). The bottles were placed in a shaking water bath at 55 °C with 130 rpm. The oxygen levels were measured by a gas chromatograph (SP 6890, Shandong Lunan Inc., China) every 4 h until the oxygen was consumed completely.

2.3. Batch anaerobic digestion tests

Batch mesophilic (37 °C) anaerobic digestion tests was performed in duplicates, the bottles after thermophilic microaerobic pretreatment were added with another 22 ml inocula to reach a relation $TS_{inoculum}/TS_{substrate}$ of 0.5, and deionized water was added to reach a total volume of 0.2 L. Another 2 bottles (marked as A) were also employed to find out the biogas yield from corn straw without pretreatment. The bottles were placed in a shaking water bath at 37 °C with 130 rpm.

2.4. Secondary thermophilic microaerobic treatment (STMT)

STMT of fermentation system was carried out when obvious decrease in daily methane yield was appeared. The VS removal efficiency before STMT was tested according to R0. During STMT, oxygen at atmospheric pressure was injected to RS1, RS2 and RS3 with a syringe to reach the oxygen loads of 5, 10 and 20 ml/g VS (VS of residual substrate). Then, the bottles were placed in a shaking water bath at 55 °C with 130 rpm for secondary thermophilic microaerobic treatment. After STMT, another 22 ml inocula were added into the bottles and flushed with N_2 for 5 min to replace the air, and then the bottles were closed with rubber stoppers. At last, the bottles were placed in shaking water bath at 37 °C with 130 rpm to continue fermentation.

2.5. Analytical methods

The biogas yield was measured by water displacement method. Biogas composition was measured by a gas chromatograph (SP 6890, Shandong Lunan Inc., China), equipped with a Porapak Q stainless steel column (180 cm long, 3 mm outer diameter) and a thermal conductivity detector. The temperatures of the injector, detector and oven were 50, 100 and 100 °C, respectively. The carrier gas was argon. SCOD (Soluble Chemical Oxygen Demand) was measured after centrifuging at 10,000 rpm for 5 min and filtering samples through a 0.45 μm glass microfiber filter and determined by spectrophotometric method (SEPA, 2002). The element contents of corn straw were detected with an elemental analyzer (Vario EL Cube). TS, VS were determined according to standard methods (APHA, 2006). Cellulose, hemicelluloses and lignin were determined according to the method as Goering and van Soest (1970) described.

3. Results and discussion

3.1. Biogas and methane productions

The biogas and methane yields during anaerobic digestion of corn straw were shown in Fig. 1. Daily methane yield increased sharply at the initial stage of anaerobic digestion and reached the maximum at the 9th day of AD. The maximum daily methane yield of thermophilic microaerobic pretreated corn straw was about 57.0% higher than that of untreated sample. The methane production peaks of thermophilic microaerobic pretreated corn straw were between 4th and 18th day of AD. Comparatively, the methane production peak of untreated corn straw was between 9th and 17th day of AD, which was lower and shorter. These means the methane yield of thermophilic microaerobic pretreated corn straw was quick and concentrated. In other words, the AD of thermophilic microaerobic pretreated corn straw was more efficient.

STMT was carried out at the 22th day of AD, when the daily methane yield in thermophilic microaerobic pretreated corn straw was obviously decreased. The VS removal efficiency at that time was determined by R0, which was 41.3%. Then 15, 30, 60 ml

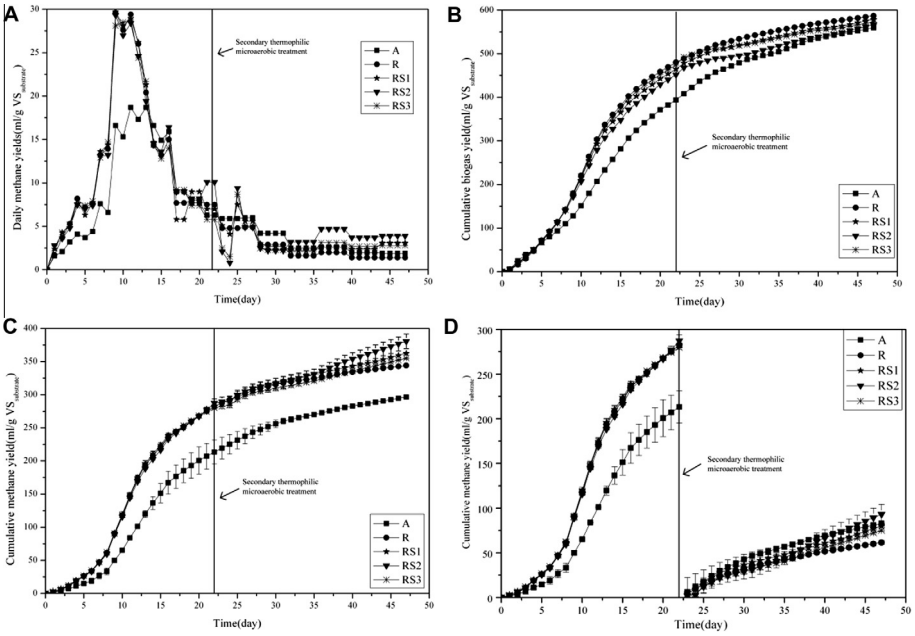


Fig. 1. The biogas and methane yields during the anaerobic digestion of corn straw (A: daily methane yield; B: cumulative biogas yield; C: cumulative methane yield; D: respective cumulative methane yield of two stage).

oxygen at atmospheric pressure was respectively added to RS1, RS2, RS3 with a syringe to reach the oxygen loads of 5, 10, 20 ml/g VS (VS of residual substrate) during secondary thermophilic microaerobic treatment. The cumulative methane yields were shown in Fig. 1C. The maximum cumulative methane yield was 380.6 ml/g $VS_{substrate}$, which was obtained from RS2 and followed by RS1, RS3, R and A. The methane yield of RS2 was 28.45% higher than that of untreated sample, the methane yield of R was 16.14% higher than that of untreated sample which was quite accordance with the result reported in previous study (Fu et al., 2015). The results in this study were quite different with that reported by Mshandete et al. (2005). In their study, a second aerobic pulse in the AD of sisal pulp gave a lower methane yield (94%) than the one with only once aerobic pretreatment. In this study, secondary thermophilic microaerobic treatment gave a 10.61% higher methane yield than that with only once thermophilic microaerobic pretreatment. The reason for reduction of methane yield in the study of Mshandete et al. (2005) may be the activity of methanogens was destroyed during secondary aeration treatment and there was no reinoculation after secondary aeration treatment. In addition, because the amount of oxygen during secondary aeration treatment was not accurately controlled, the excess organic substrate consumed may be another reason for the decrease in methane yield.

The cumulative methane yields before and after secondary thermophilic microaerobic treatment were shown in Fig. 1D. The cumulative methane yield of thermophilic microaerobic pretreated corn straw before STMT was about 283.0 ml/g $VS_{substrate}$, which was 32.68% higher than that of untreated sample. As for the cumulative methane yield after STMT, the maximum methane yield was obtained when the oxygen loads during STMT was 10 ml/g VS (VS of residual substrate), which was 93.3 ml/g $VS_{substrate}$ and 12.88% higher than that of untreated sample. In addition, the cumulative methane yields of untreated sample and RS2 after STMT were respectively 34.74% and 51.46% higher than that of once thermophilic microaerobic pretreated sample, which was only 61.6 ml/g $VS_{substrate}$.

The cumulative methane yields of RS1 and RS3 after STMT were also higher than that of once thermophilic microaerobic pretreated sample, which meant the STMT process was surely improved the methane yield of corn straw during the late stage of anaerobic digestion. However, the increase in oxygen load during STMT did not lead to high methane yield, which demonstrated once again the importance of oxygen loads during microaerobic treatment. On the one hand, the excessive oxygen may inhibit the activity of methanogens and decrease the methane yield (Xu et al., 2014). In addition, facultative organisms have higher growth rates and would out-compete strict anaerobes under high oxygen levels due to substrate competition (Lim and Wang, 2013). On the other hand, insufficient oxygen will not be strong enough to support the growth of facultative organisms.

3.2. SCOD variations during anaerobic digestion

Soluble COD is a parameter that represents the extent of solubilization. Pretreatment methods that disintegrate the substrate before digestion to improve the biogas yield during the subsequent digestion are most commonly characterized in terms of solubilization of COD (Appels et al., 2011). The variation of SCOD during anaerobic digestion of corn straw was shown in Fig. 2. The average SCOD of thermophilic microaerobic pretreated corn straw at the first day of anaerobic digestion was 7962.5 mg/L, which was 26.39% higher than that of untreated sample. The SCOD concentrations of thermophilic microaerobic pretreated corn straw increased sharply and reached the peak firstly. Comparatively, the SCOD concentrations peak of untreated sample came later. The earlier SCOD concentrations peak in thermophilic microaerobic pretreated corn straw means the fermentation structural and compositional barriers of corn straw were disrupted partly during TMP.

The SCOD concentration after secondary thermophilic microaerobic treatment obviously increased, which could attribute to the high hydrolysis under microaerobic condition. According to Fig. 1A, the higher SCOD concentration in RS1, RS2 and RS3 also resulted to the increase in daily methane yield. Increased SCOD concentration under microaerobic condition was also reported by Mshandete et al. (2005) and Jagadabhi et al. (2010), the reason behind this may be the higher hydrolysis rate under microaerobic condition.

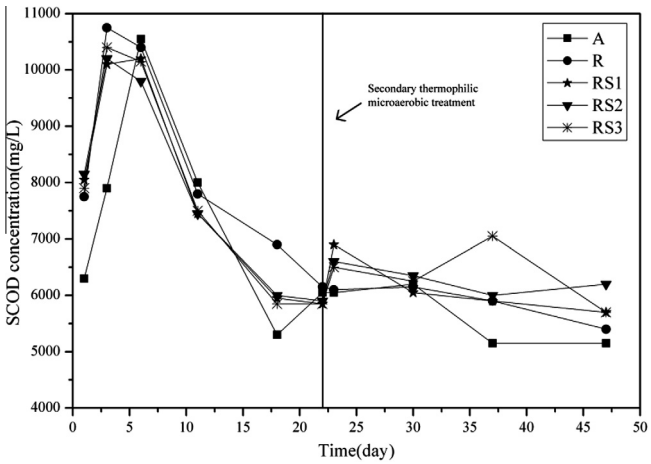


Fig. 2. The variation of SCOD during anaerobic digestion of corn straw.

3.3. VS removal efficiency

During the digestion process, VS are degraded to a certain extent and converted into biogas and the degree of stabilization is often expressed as the percent reduction in VS (Appels et al., 2008). In this study, the VS removal efficiency was calculated as following:

$$\text{VS removal efficiency} = \frac{\text{initial VS} - \text{remaining VS}}{\text{initial VS}} \quad (1)$$

where initial VS is the VS of substrate added, remaining VS is calculated by the remaining total VS in the fermentation bottles minus the VS of blank (only inoculum).

The VS removal efficiencies of A, R, RS1, RS2 and RS3 were 63.24 ± 0.15%, 69.82 ± 1.94%, 81.09 ± 0.90%, 81.85 ± 0.94% and 79.54 ± 0.94%, respectively. The VS removal efficiency of once thermophilic microaerobic pretreated sample was 10.40% higher than that of untreated sample. The maximum VS removal efficiency was obtained from the sample that the oxygen loads was 10 ml/g VS (VS of residual substrate) during the secondary thermophilic microaerobic treatment. It was 29.43% and 17.23% higher than those of untreated and once thermophilic microaerobic pretreated samples, respectively. The higher VS removal efficiency means more substrate was digested in anaerobic digestion, which would be better for the reduction of fermentation residue.

4. Conclusions

Secondary thermophilic microaerobic treatment during the batch anaerobic digestion of corn straw obviously improves the methane yield and VS removal efficiency. The maximum methane yield was obtained when the oxygen loads during STMT was 10 ml/g VS (VS of residual substrate), which was respectively 28.45% and 10.61% higher than those of untreated and once thermophilic microaerobic pretreated samples. The maximum VS removal efficiency was also obtained when the oxygen loads during STMT was 10 ml/g VS (VS of residual substrate), which was respectively 29.43% and 17.23% higher than those of untreated and once thermophilic microaerobic pretreated samples.

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